

Regulatory architecture of H3K4me1 and H3K4me3 peaks

We quantified the regulatory context of H3K4me3 and H3K4me1 peaks in C1 by intersecting peak sets with ENCODE candidate cis-regulatory elements (cCREs) and gene-model features. cCRE overlaps indicate a promoter-centric architecture for H3K4me3: promoter-like sequences (PLS), together with proximal enhancer-like signatures (pELS), account for the vast majority of overlaps, whereas distal enhancer-like signatures (dELS) are minimal (Figure 1). In contrast, H3K4me1 peaks are enhancer-oriented, with dELS comprising the largest component and pELS also prominent; PLS contributes a comparatively smaller fraction. Overlaps with CTCF-bound and other non-enhancer/non-promoter cCRE classes are negligible for both marks.

Gene-model distributions independently corroborate these patterns (Figure 2). H3K4me3 peaks are concentrated at annotated promoters, with secondary representation in exons and introns and minimal signal in UTRs or other categories. H3K4me1 peaks are dominated by intronic overlaps, with a smaller promoter fraction and limited exonic/UTR representation; an “other” category is observed only for H3K4me1. The intronic enrichment of H3K4me1 is consistent with its preferential intersection with dELS, which frequently localize within introns, whereas the strong promoter enrichment of H3K4me3 aligns with its predominant overlap with PLS.

Together, the cCRE-based classification and gene-model overlaps provide concordant evidence that, in C1, H3K4me3 is promoter-biased, whereas H3K4me1 is enhancer-biased with marked intronic localization.

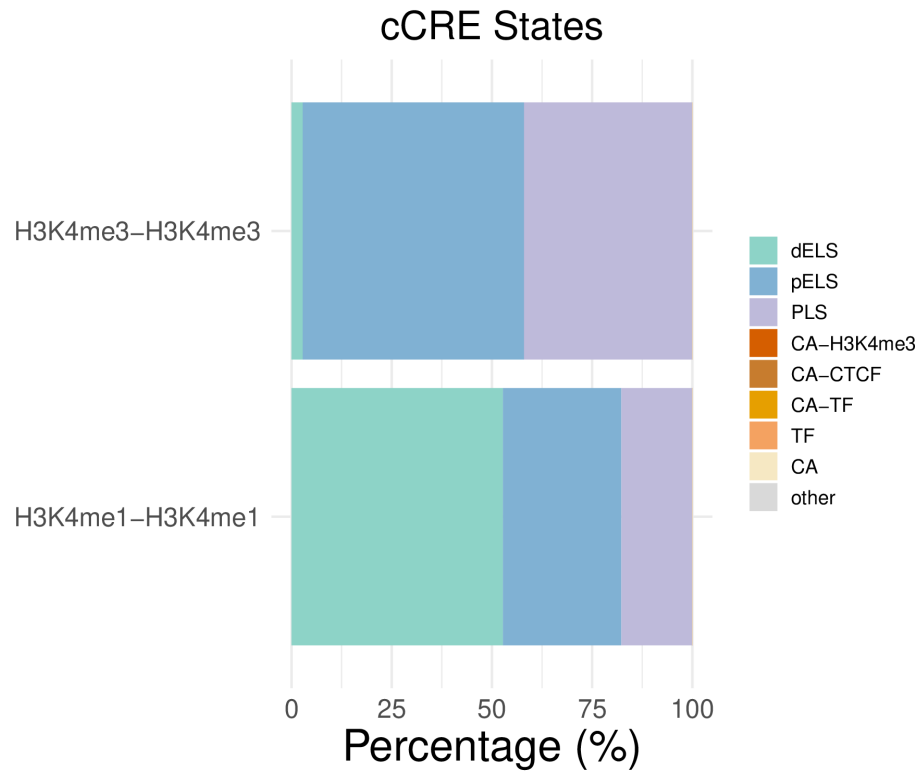


Figure 1. cCRE class distribution for H3K4me3 and H3K4me1 peaks. Peak coordinates were intersected with cCRE annotations to assign regulatory classes. H3K4me3 peaks overlap mainly promoter-like (PLS) and proximal enhancer-like (pELS) elements, whereas H3K4me1 peaks are enriched for distal enhancers (dELS) with smaller promoter-like overlap.

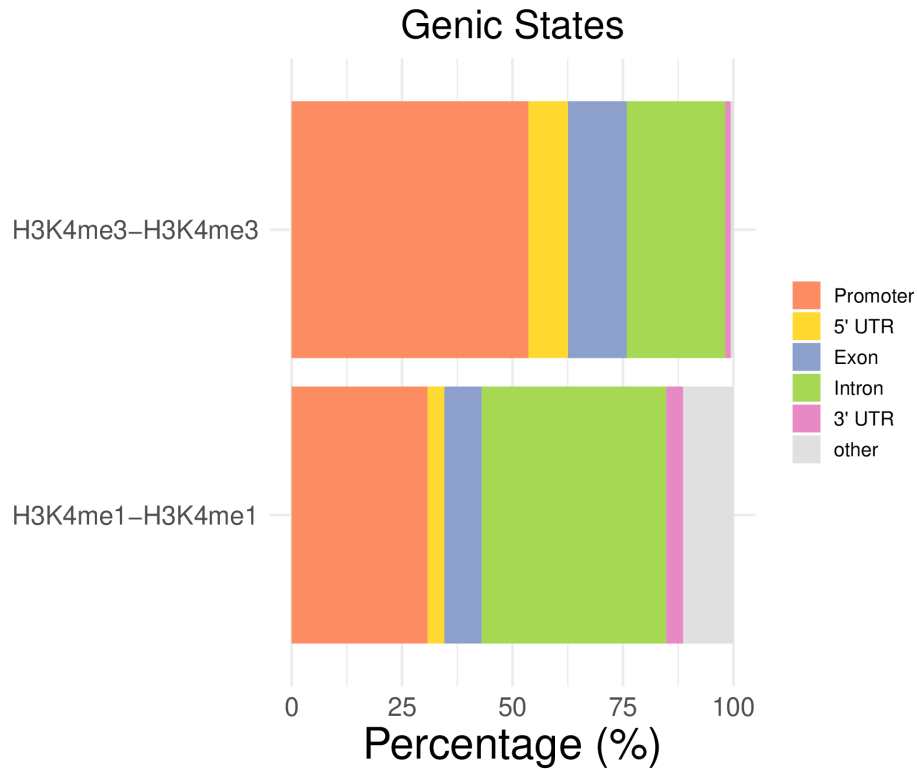


Figure 2. Genic distribution of peaks for H3K4me3 and H3K4me1. Peaks were assigned to gene features and summarized for the H3K4me3–H3K4me3 and H3K4me1–H3K4me1 datasets. H3K4me3 peaks concentrate at promoters, whereas H3K4me1 peaks are predominantly intronic with an additional 'other' fraction; exons and UTRs contribute smaller shares.

Quantitative concordance between H3K4me1 and H3K4me3 across shared peaks

Per-peak comparison of average signal intensities across the 1,546 union peaks revealed a pronounced asymmetry in dynamic range between the marks (Figure 3). H3K4me3 spanned from near 0 to >35, whereas H3K4me1 was comparatively compressed, clustering around 4–8 with a lower bound near 4–5. The joint distribution was dispersed with little positive trend, featuring (i) a vertical stack at H3K4me1 ~4–6 that spans H3K4me3 ~10–30+, and (ii) a low-H3K4me3 band (~0–4) concentrated at intermediate H3K4me1 (~7–12). These features indicate heteroscedasticity, with H3K4me3 contributing most of the dynamic range across shared loci and H3K4me1 exhibiting a comparatively stable baseline.

Consistent with this structure, cross-mark correlations were weakly negative (Figure 4). The H3K4me1–H3K4me3 Pearson correlation was -0.28 and the Spearman correlation was -0.12, whereas self-correlations were 1.00. The stronger attenuation of the rank-based metric toward zero indicates minimal monotonic association, while the more negative Pearson value reflects a modest linear anticorrelation driven by the wider variance of H3K4me3.

Notable outliers and subpopulations further support heterogeneous regulatory states among shared peaks (Figure 3). Peaks with very high H3K4me3 (>35) but only moderate H3K4me1 ($\sim 5\text{--}9$) are consistent with promoter-like activity, whereas peaks with elevated H3K4me1 (>15) and low H3K4me3 (<10) are compatible with enhancer-like profiles. The intermediate-H3K4me1, low-H3K4me3 band suggests an enhancer-skewed cohort, while the vertical stack at low-to-moderate H3K4me1 coupled to widely varying H3K4me3 implies an H3K4me1 baseline superimposed on promoter-associated H3K4me3 variation.

Together, these patterns explain the weak anticorrelation and limited overall concordance between the marks across shared loci.

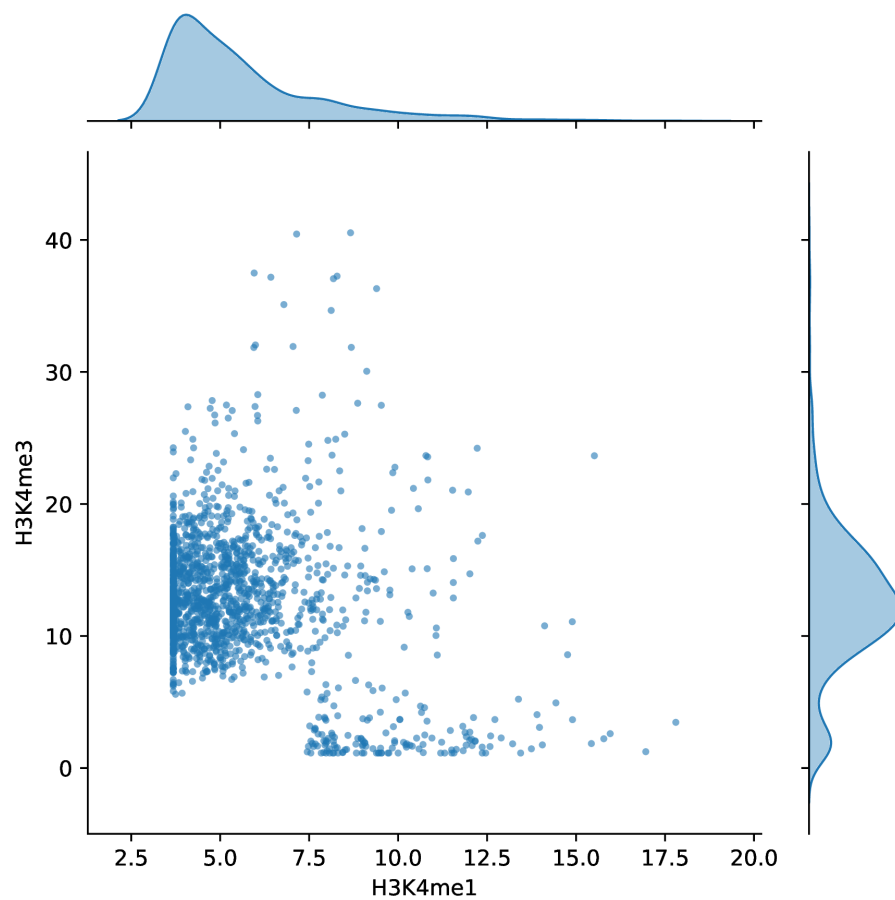


Figure 3. Per-peak H3K4me1 versus H3K4me3 signal comparison. Each point is a union peak with averaged coverage for H3K4me1 (x) and H3K4me3 (y). H3K4me1 concentrates around 4–8, whereas H3K4me3 spans a wider range with a low group near 0–4 and a main spread around 10–20+, including outliers at very high H3K4me3 or high H3K4me1 with low H3K4me3.

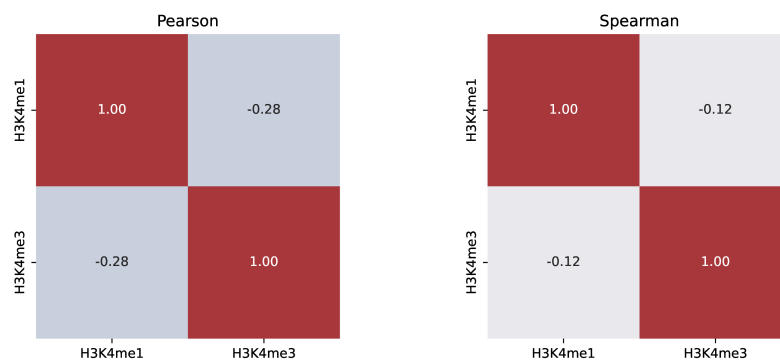


Figure 4. H3K4me1 and H3K4me3 show weak anticorrelation across peaks. Pearson and Spearman correlation heatmaps summarize pairwise associations across the union peak set. Self-correlations are 1.00, while the cross-mark correlation is negative (Pearson -0.28; Spearman -0.12), indicating lower concordance in both linear and rank order.

Transcription factor motif enrichment across detected peaks

Genome-wide motif analysis revealed a global skew toward overrepresentation of transcription factor (TF) motifs across the detected peak sets. The enrichment heatmap (Figure 5) is dominated by positive values, with only sparse, isolated instances of weak or negative enrichment, indicating broad enrichment for TF-binding signatures relative to background and consistent capture of functional regulatory sequence.

Clustering of enrichment profiles identified subsets of TFs and peak groups with shared preferences (Figure 5). Intermittent, high-intensity foci arise against a background of pervasive, moderate enrichment, suggesting that while many TFs are generally overrepresented, distinct TF modules drive particularly strong enrichment within specific peak subsets.

Relating these patterns to the regulatory architectures established above, promoter-biased peak sets exhibit broader, moderate-level enrichment across many TFs, consistent with the dense, multifactorial binding environment characteristic of promoters. In contrast, enhancer-biased peak sets display comparatively narrower but higher-magnitude enrichment concentrated in specific TF modules, consistent with more selective, context-dependent enhancer grammars. The scattered distribution of intense signals without strong opposing gradients supports an interleaved organization in which enhancer-focused TF modules are superimposed on a generally enriched landscape. Overall, the breadth and

magnitude of motif overrepresentation corroborate the coexistence of promoter- and enhancer-driven regulatory programs within the detected peaks (Figure 5).

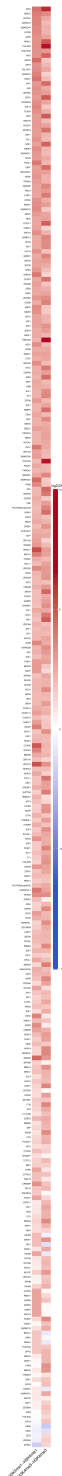


Figure 5. Transcription factor motif enrichment across detected peaks. Heatmap summarizing normalized enrichment for numerous TF binding motifs within the profiled peak set. Most factors show positive enrichment (red), with only a few near-neutral or weakly depleted signals (white to blue).

Functional pathway enrichment of C1 peak-linked genes highlights p53 and DNA repair

Over-representation analysis of genes linked to C1 peaks revealed a sharply polarized enrichment landscape dominated by DNA damage and checkpoint programs (Figure 6). The strongest signals mapped to P53_PATHWAY and DNA_REPAIR, with additional but comparatively weaker enrichment in MYC_TARGETS_V1, E2F_TARGETS, G2M_CHECKPOINT, MTORC1_SIGNALING, and HEME_METABOLISM. In contrast, many other functional categories (for example, oxidative phosphorylation, interferon responses, apoptosis, and diverse metabolic pathways) showed little or no enrichment, indicating that C1 peak-linked genes are selectively concentrated in genome-stability and cell-cycle regulatory modules.

Mark-specific stratification showed that these pathway signals were driven almost entirely by promoter-proximal H3K4me3-retaining loci (H3K4me3→H3K4me3), whereas enhancer-skewed regions marked only by H3K4me1 (H3K4me1→H3K4me1) contributed minimally (Figure 6). This pattern is consistent with the canonical interpretation of CUT&Tag-derived histone landscapes, in which H3K4me3 marks active promoters tightly coupled to pathway membership, while H3K4me1 predominantly labels distal enhancers with more context-specific outputs (Shi et al., 2025). The co-enrichment of p53 signaling and DNA repair with MYC/E2F cell-cycle and mTORC1-related programs suggests concurrent engagement of checkpoint/repair axes and pro-proliferative/growth-control circuitry within the promoter-anchored gene set, a configuration aligned with stress-response biology in which p53-mediated surveillance interfaces with DNA repair and cell-cycle control (Wen et al., 2026).

Together, these data indicate that C1 peaks preferentially tag promoter-proximal regulatory elements of genes governing p53 signaling and DNA repair, with secondary contributions

from MYC/E2F-driven cell-cycle and mTORC1 pathways, while enhancer-only loci show little systematic pathway aggregation.

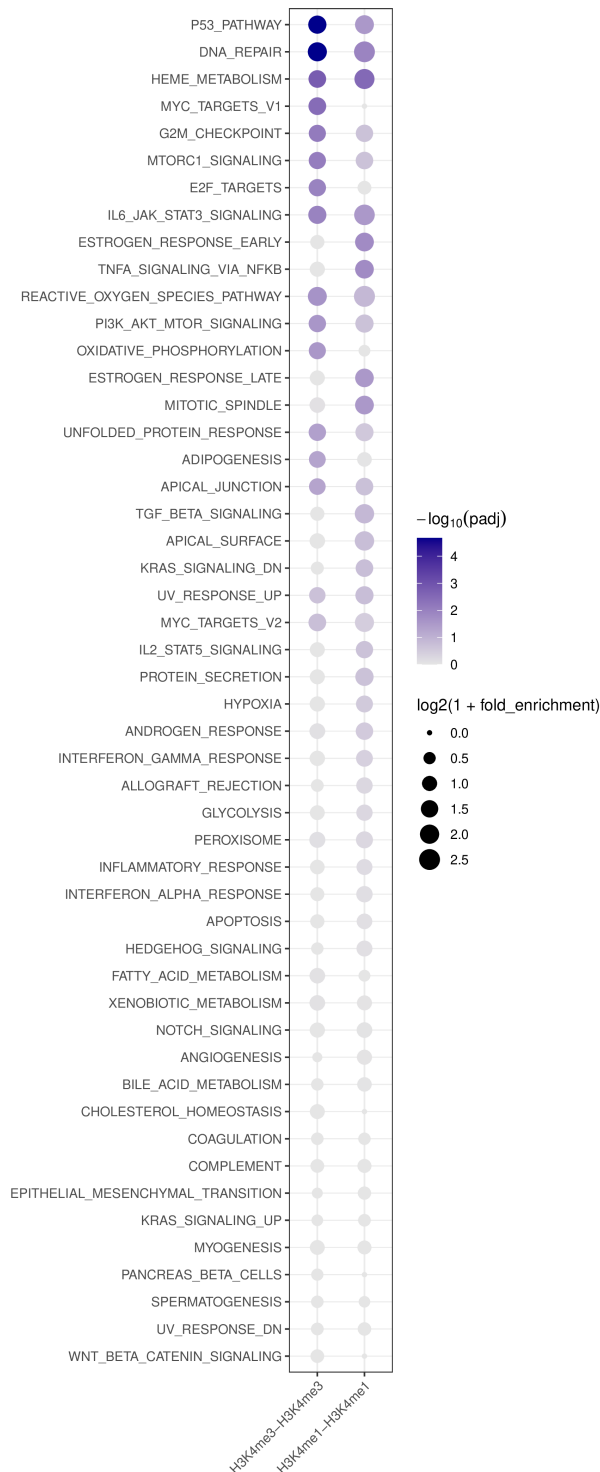


Figure 6. C1 peak-linked genes enrich for p53 and DNA repair. Pathway enrichment of genes associated with C1 peaks highlights prominent signals in P53 pathway, DNA repair, MYC targets, G2M checkpoint, and MTORC1 signaling. Signals are concentrated in the H3K4me3→H3K4me3 group, while the H3K4me1→H3K4me1 group shows little enrichment.

References

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